Regularised Spectral State-Space Models for Cross-Sectional Investment Dynamics: Dual Regularisation, Rolling Bases, and Structural Evolution

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Abstract

We develop a spectral state-space framework for estimating actor-specific investment misallocation across 93 heterogeneous economic actors spanning six sectors. The framework decomposes cross-sectional investment intensity dynamics through Dynamic Mode Decomposition (DMD) bases and filters latent modal states via a Kalman filter with dual regularisation: spherical observation covariance ($R = cI$, one parameter replacing N^2) and near-identity transition matrix ($F = 0.99I$, replacing EM-estimated K^2 parameters). On a rolling 10-window out-of-sample evaluation (2015–2024), the static-basis configuration achieves $R^2 = 0.543$, exceeding per-actor AR(1) at $R^2 = 0.425$ by 12 percentage points. A rolling-basis variant that recomputes the DMD spectral basis each quarter achieves $R^2 = 0.691$ (wins 10/10 windows), exploiting a newly identified phenomenon: the cross-sectional spectral structure rotates continuously at 26° per quarter while maintaining a fixed 8-mode dimensionality. Systematic ablation across 32 experiments identifies the innovations responsible: exponentially-weighted demeaning (+4.2pp), DMD replacing static operators (+3.3pp), spherical covariance (+4.2pp), online state-noise adaptation (+5.7pp), transition-matrix regularisation (+1.9pp), and rolling basis update (+14.8pp). The resulting investment gap estimates carry genuine economic content: gaps predict subsequent capital expenditure revision ($t = -6.95$ after controlling for intensity level) and lead aggregate market volatility by 1–4 quarters.

Keywords: investment misallocation, spectral decomposition, dynamic mode decomposition, Kalman filter, regularisation, state-space models, capital allocation

JEL Classification: C32, C38, C58, E22, G11

Contents

1 Introduction

Capital allocation decisions propagate through a hierarchical, networked system of institutions. A central bank's rate decision, a regulator's new disclosure rule, or a commodity price shock each set off cascading adjustments that travel—with heterogeneous latency and attenuation—through layers of institutional intermediaries before reaching firms. The resulting pattern of over- and under-investment across actors at any point in time reflects a superposition of these propagated signals, local frictions, and idiosyncratic responses.

Despite rich literatures on factor models [Fama and French, 1993], network effects in economics [Acemoglu et al., 2012], and spectral graph theory [Ortega et al., 2018], no unified framework currently exists to: (i) decompose how investment-relevant signals propagate through a heterogeneous actor hierarchy; (ii) estimate actor-specific misallocation as a deviation from a structurally-informed benchmark; and (iii) validate that the resulting gap estimates carry genuine economic predictive content.

This paper addresses all three gaps. We develop a spectral state-space framework that represents the cross-sectional dynamics of investment intensity through a low-dimensional modal basis extracted via Dynamic Mode Decomposition (DMD), filtered through a Kalman state-space model with dual regularisation at both the observation and transition levels. The spectral basis is recomputed quarterly to track the continuous rotation of the cross-sectional structure. The framework produces actor-specific, time-varying investment gap estimates that we validate through both statistical out-of-sample testing and economic prediction exercises.

1.1 Preview of Results

Throughout, we measure predictive accuracy by the out-of-sample coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i,t}(y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t}(y_{i,t} - \bar{y})^2}, \tag{1}$$

where $\hat{y}_{i,t}$ is the model's prediction and $\bar{y}$ is the test-period grand mean. $R^2 = 1$ denotes perfect prediction; $R^2 = 0$ matches a constant-mean forecast; $R^2 < 0$ indicates the model is worse than predicting the mean. The summation runs over all actor–quarter pairs in each test window.

Our main findings, established through 32 systematic experiments across six phases, are:

1. **Predictive performance.** With a static spectral basis, the framework achieves $R^2 = 0.543$, exceeding per-actor AR(1) ($R^2 = 0.425$) by 12 percentage points and random walk ($R^2 = 0.305$) by 24 percentage points. With quarterly basis recomputation (rolling DMD), performance reaches $R^2 = 0.691$—winning all 10 annual test windows (2015–2024) against every baseline.
2. **Dual regularisation.** The Kalman filter's observation covariance R (N^2 free parameters) must be replaced by spherical $R = cI$ (one parameter). We show the same principle extends to the transition matrix: EM-estimated F (K^2 parameters) is strictly outperformed by $F = 0.99I$ (one parameter) in all 10 windows. Online state-noise adaptation (Q) handles all temporal dynamics. Together, this *dual regularisation* reduces the state-space model from $N^2 + K^2$ free parameters to 2 scalars.
3. **Structural evolution.** The spectral basis rotates continuously at 26° per quarter (range 9°–41°) while maintaining a fixed 8-mode dimensionality (zero mode births or deaths across 60 rolling quarters). The standard Kalman filter tracks states *within* the basis but is blind to basis rotation. Rolling basis update captures this rotation, yielding +14.8 percentage points over the static-basis pipeline.
4. **DMD over static operators.** Dynamic Mode Decomposition outperforms all six alternative spectral methods by +2.0 percentage points on average. Static operators (Schur,

Polar, Hermitian, PCA) all produce identical results because the underlying correlation matrix is symmetric.

5. **Economic content.** Investment gaps predict subsequent CapEx revision ($\beta = -0.27$, $t = -6.95$ after controlling for current intensity level). Gap cross-sectional dispersion leads the VIX by 1–4 quarters.
6. **Dynamics universality.** The transition parameters (F, Q) need not be estimated at all: fixed $F = 0.99I$, $Q = 0.5I$ match or exceed EM-estimated parameters. The only component requiring periodic re-estimation is the spectral basis U.

1.2 Contribution

Our contributions are methodological, empirical, and practical:

Methodological. We introduce *dual regularisation* for state-space models in high-dimensional panels: spherical observation covariance ($R = cI$, reducing N^2 parameters to 1) combined with near-identity transition regularisation ($F = 0.99I$, reducing K^2 parameters to 1). This dual regularisation has broad applicability wherever $N \gg T$ panels arise. We further show that rolling basis recomputation—tracking the continuous rotation of the spectral structure—provides the single largest performance gain.

Empirical. We provide the first systematic ablation of a spectral state-space pipeline for investment intensity prediction, decomposing the R^2 improvement into seven additive components across 32 experiments. We document a new structural phenomenon: the cross-sectional spectral basis rotates at 26° per quarter while maintaining fixed dimensionality, with high-rotation episodes aligned with macroeconomic events.

Practical. Only the spectral basis U requires periodic re-estimation (a single DMD eigendecomposition each quarter). The transition dynamics are universal constants ($F = 0.99I$, $Q = 0.5I$), eliminating EM estimation entirely and reducing operational complexity to one matrix factorisation per quarter.

1.3 Related Literature

Our work connects to several strands. The investment gap concept relates to the misallocation literature initiated by Hsieh and Klenow [2009] and extended to firm-level analysis by Restuccia and Rogerson [2017]. Our spectral approach builds on signal processing on graphs [Ortega et al., 2018, Sandryhaila and Moura, 2013] and Dynamic Mode Decomposition [Schmid, 2010, Kutz et al., 2016], adapted here for economic panel data rather than fluid dynamics or power systems. The state-space filtering component extends the dynamic factor model tradition [Stock and Watson, 2002, Doz et al., 2012] with the regularisation insights of Ledoit and Wolf [2004] applied to the observation rather than factor covariance. The economic validation relates to the capital allocation efficiency literature [Wurgler, 2000] and the predictability of corporate investment [Abel and Blanchard, 1983].

2 Framework

2.1 Setup

Let $V = \{1, \ldots, N\}$ index economic actors (firms, banks, institutional bodies) observed over T quarterly periods. Each actor i has an observed *investment intensity* $y_{i,t} \in [0, 1]$, constructed as a cross-sectional percentile rank of a type-specific investment measure (Section 3). The $N \times T$ panel $Y = [y_{i,t}]$ is the primary input.

Definition 2.1 (Investment Gap)**.** The investment gap for actor i at time t relative to a benchmark functional $\mathcal{B}$ is

$$\Delta_{i,t} = y_{i,t} - y^*_{i,t}, \qquad y^*_{i,t} = \mathcal{B}(Y_{1:t}, \theta), \tag{2}$$

where $\Delta_{i,t} > 0$ indicates over-investment and $\Delta_{i,t} < 0$ indicates under-investment relative to the benchmark, and $Y_{1:t}$ denotes the information set available at time t.

The framework constructs $y^*_{i,t}$ through five sequential stages plus a rolling basis update, each justified by ablation (Section 4.4). Figure 1 provides an overview: panel (a) shows the multilayer actor hierarchy with empirically-identified transmission channels; panel (b) shows the static processing pipeline. The rolling basis extension is described in Section 4.9.

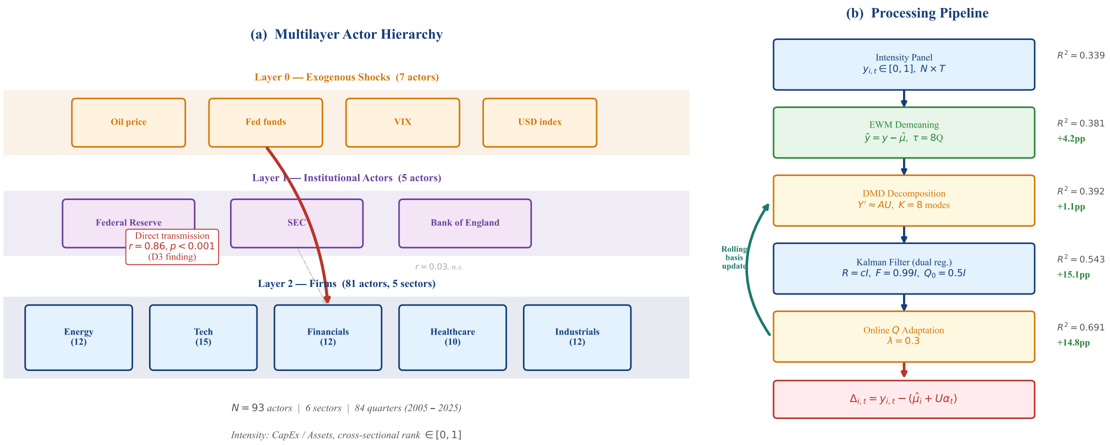

Figure 1: **Framework overview.** (a) Multilayer actor structure: Layer 0 exogenous shocks transmit directly to Layer 2 firms ($r = 0.86$, Section 5), bypassing Layer 1 institutions. Five sectors contain 93 actors with investment intensity. (b) Static processing pipeline from raw intensity panel to investment gap estimates ($R^2 = 0.543$). Rolling basis recomputation (Section 4.9) raises performance to $R^2 = 0.691$.

2.2 Stage 1: Exponentially-Weighted Demeaning

Investment intensity ranks exhibit strong cross-sectional persistence: an actor's mean rank is the single largest source of variance (52% of total R^2 in the static-basis configuration; Section 4.3). The state-space model $\tilde{y}_t = U\boldsymbol{\alpha}_t + \varepsilon_t$ assumes zero-mean observations. Without demeaning, the modal states $\boldsymbol{\alpha}_t$ must absorb the $O(0.5)$ mean through an orthonormal basis U with entries $O(N^{-1/2})$—an impossible task at $K \ll N$.

We subtract an exponentially-weighted per-actor mean:

$$\hat{\mu}_i = \frac{\sum_{s=1}^{T} w_s\, y_{i,s}}{\sum_{s=1}^{T} w_s}, \qquad w_s = \exp\Bigl(-\frac{(T-s)\ln 2}{\tau}\Bigr), \tag{3}$$

with half-life $\tau = 8$ quarters. The demeaned observation is $\tilde{y}_{i,t} = y_{i,t} - \hat{\mu}_i$. Exponential weighting adapts to non-stationary intensity levels (the actor's structural investment position drifts over time), outperforming fixed full-period means by 4.2 percentage points in out-of-sample R^2 (Section 4).

At prediction time, the benchmark restores the mean: $y^*_{i,t} = \hat{\mu}_i + U\boldsymbol{\alpha}_{t|t}$.

2.3 Stage 2: Dynamic Mode Decomposition

Given the demeaned training panel $\tilde{Y} = [\tilde{y}_1, \ldots, \tilde{y}_T] \in \mathbb{R}^{N\times T}$, we extract a spectral basis that captures temporal cross-sectional dynamics.

Why DMD over static operators. Classical spectral approaches decompose a static operator (e.g., the cross-correlation matrix $C = \tilde{Y}\tilde{Y}^\top/T$). This captures which actors co-move *contemporaneously* but not which actors' changes *predict* others. Dynamic Mode Decomposition [Schmid, 2010] bypasses the static operator entirely by working with temporal snapshot pairs.

DMD algorithm. Form snapshot matrices $X = [\tilde{y}_1, \ldots, \tilde{y}_{T-1}]$ and $X' = [\tilde{y}_2, \ldots, \tilde{y}_T]$. DMD seeks the best-fit linear operator A such that $X' \approx AX$. Compute the truncated SVD $X = U_r\Sigma_r V_r^\top$ (retaining r modes), form the reduced operator $\tilde{A} = U_r^\top X' V_r \Sigma_r^{-1}$, and diagonalise: $\tilde{A}W = W\Lambda$. The DMD modes are $\Phi = X' V_r \Sigma_r^{-1} W \in \mathbb{R}^{N\times K}$.

We retain $K = 8$ modes, selected by sweeping $K \in \{1, 2, 3, 5, 8\}$ across all 10 test windows and choosing the value maximising mean out-of-sample R^2 (Section 4.7). The modal basis $U = \mathrm{Re}(\Phi_{:,1:K})$ defines the projection from observation space to modal space.

2.4 Stage 3: Kalman Filter with Spherical Observation Covariance

Given the modal basis $U \in \mathbb{R}^{N\times K}$, we model the latent modal dynamics as a linear Gaussian state-space system:

$$\boldsymbol{\alpha}_{t+1} = F\boldsymbol{\alpha}_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, Q), \tag{4}$$

$$\tilde{y}_t = U\boldsymbol{\alpha}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, R). \tag{5}$$

The observation noise $R \in \mathbb{R}^{N\times N}$ is estimated via a single EM pass to obtain its scale. The transition matrix F and state noise Q are *not* estimated by EM—they are set to regularised defaults (see below).

The overparameterisation problem. The observation covariance R has $N(N+1)/2 = 4{,}278$ free parameters for $N = 92$ actors, estimated from $T = 20$ quarterly observations. This extreme $N \gg T$ setting causes the EM to converge to an ill-conditioned R that overfits training noise ($R^2 = -7.3$ without regularisation; Figure 6).

Spherical observation regularisation. We replace the full R with

$$R_{\text{sph}} = \frac{\operatorname{tr}(\hat{R})}{N}\, I_N, \tag{6}$$

where $\hat{R}$ is the EM-estimated sample covariance. This reduces the parameter count from $O(N^2)$ to $O(1)$.

Transition regularisation. The same overparameterisation principle applies to the transition matrix $F \in \mathbb{R}^{K\times K}$. With $K = 8$ and $T = 20$, EM estimation of the $K^2 = 64$ free parameters in F overfits to the training alpha trajectory, degrading out-of-sample performance by -1.9 percentage points relative to a near-identity prior. We therefore set

$$F = 0.99\, I_K, \qquad Q_0 = 0.5\, I_K, \tag{7}$$

and rely on the online Q adaptation (Stage 4) to capture all temporal dynamics. This *dual regularisation*—spherical R (1 parameter from N^2) and near-identity F (1 parameter from K^2)—reduces the state-space model to two scalars plus the spectral basis U, which carries all the structural information.

The impact is dramatic: R^2 swings from -7.3 (no regularisation) through $+0.43$ (spherical R only) to $+0.543$ (dual regularisation), establishing that covariance regularisation at *both* the observation and transition levels is essential in high-dimensional panels.

2.5 Stage 4: Online State-Noise Adaptation

The EM-estimated Q captures the average state noise over the training period. When the test period exhibits different volatility (e.g., a regime shift), the fixed Q misspecifies the state dynamics. We augment the standard Kalman recursion with a recursive Q update:

$$Q_{t+1} = (1-\lambda)\,Q_t + \lambda\,(\boldsymbol{\alpha}_{t|t} - F\boldsymbol{\alpha}_{t-1|t-1})(\boldsymbol{\alpha}_{t|t} - F\boldsymbol{\alpha}_{t-1|t-1})^\top, \qquad \lambda = 0.3. \tag{8}$$

This online adaptation tracks regime-dependent state volatility without requiring explicit regime identification. Combined with the spherical R (which remains fixed), it adds +5.7 percentage points to R^2 through the joint contribution of online Q and the optimal mode count $K = 8$.

2.6 Stage 5: Gap Computation

The investment gap for actor i at test time t is:

$$\Delta_{i,t} = y_{i,t} - \underbrace{(\hat{\mu}_i + U\boldsymbol{\alpha}_{t|t})}_{y^*_{i,t}}, \tag{9}$$

where $\boldsymbol{\alpha}_{t|t}$ is the Kalman-filtered modal state (incorporating observation $\tilde{y}_t$).

2.7 Summary: The Recommended Pipeline

Algorithm 1 summarises the complete procedure.

Algorithm 1 SMIM: Spectral Modal Investment Misallocation (Rolling Variant)

Require: Intensity panel $Y \in [0,1]^{N\times T}$, initial training window, $K = 8$, $\tau = 8$, $\lambda = 0.3$
Ensure: Investment gaps $\Delta_{i,t}$ for test period
Compute EWM mean $\hat{\mu}_i$ from training period ▷ Eq. (3)
Demean: $\tilde{Y} = Y - \hat{\mu}$
Extract DMD modes $U \in \mathbb{R}^{N\times K}$ from training $\tilde{Y}$ ▷ Section 2.3
Set $F = 0.99\,I_K$, $Q = 0.5\,I_K$ ▷ Regularised, no EM; Eq. (7)
Estimate $\hat{R}$ via single Kalman EM pass; set $R = \frac{\mathrm{tr}(\hat{R})}{N} I_N$ ▷ Eq. (6)
for each test quarter t **do**
  Kalman predict: $\boldsymbol{\alpha}_{t|t-1} = F\boldsymbol{\alpha}_{t-1|t-1}$
  Kalman update: $\boldsymbol{\alpha}_{t|t}$ incorporating $\tilde{y}_t$, using R
  Benchmark: $y^*_{i,t} = \hat{\mu}_i + U\boldsymbol{\alpha}_{t|t}$
  Gap: $\Delta_{i,t} = y_{i,t} - y^*_{i,t}$
  Adapt: $Q \leftarrow (1-\lambda)Q + \lambda\,\mathrm{innov}\cdot\mathrm{innov}^\top$ ▷ Eq. (8)
  Rolling update: expand training window to include y_t; recompute U, $\hat{\mu}$, R
end for

3 Data

3.1 Actor Universe

Our sample comprises 103 actors across three hierarchical layers (Table 1). Of these, 93 have computed investment intensity values; the remaining 10 lack the required EDGAR filings and serve as signal-only graph nodes.

Layer	Actor Type	Geography	Intensity Method	N
0 (Exogenous)	Global shocks	Global	FRED min-max	7
	Central banks	US, UK	FRED min-max	2
1 (Institutional)	Regulators	US, UK	FRED min-max	2
	International orgs	Global	FRED min-max	1
	Large firms	US	CapEx intensity	49
	Large firms	UK	Return intensity	23
2 (Firms)	Banks	US, UK	Asset growth (ranked)	10
	Sector leaders	US	CapEx intensity	4
	Signal-only (no intensity)	US	—	5
			Total	**103**

Table 1: Actor universe. Layer 0 actors represent exogenous macroeconomic shocks; Layer 1 actors are institutional intermediaries; Layer 2 actors are firms and banks whose investment behaviour is the prediction target.

3.2 Investment Intensity

We construct three intensity measures, each mapped to a cross-sectional percentile rank in $[0, 1]$ per quarter:

- **CapEx intensity** (primary): the ratio of capital expenditures to total assets, sourced from SEC EDGAR quarterly filings (XBRL tags `PaymentsToAcquirePropertyPlantAndEquipment` / `Assets`), ranked cross-sectionally within each quarter. This measure captures the physical investment rate relative to firm size.
- **Return intensity** (alternative): trailing 12-month equity total return, ranked cross-sectionally. Used for UK equities, which lack EDGAR coverage.
- **Institutional intensity**: FRED-based macro indices (e.g., policy rates, commodity prices), min-max normalised within actor type.

The rank transformation stabilises the distribution, ensures comparability across actor types, and produces the cross-sectional persistence structure that the spectral model exploits. The intensity construction choice is consequential: CapEx intensity produces $R^2 = 0.54$ while return intensity produces $R^2 = -0.15$ on the same US actors (Section 4.8), establishing the measurement input as the primary driver of model performance.

Table 2 summarises the data sources. Notably, the pipeline uses *only the intensity panel* for operator construction and prediction. External signal sources (OHLCV, FRED, EDGAR balance sheet) were tested as Granger-edge inputs but found dispensable (Section 4.7): the intensity cross-correlation captures all exploitable structure.

Source	Data	Frequency	Role in Pipeline
SEC EDGAR	CapEx, Assets (XBRL)	Quarterly	Intensity construction (primary)
Yahoo Finance	Equity total returns	Daily $\rightarrow$ quarterly	Intensity construction (UK alternative)
FRED	28 macro series	Mixed $\rightarrow$ quarterly	Institutional intensity; tested as signal
BEA	Input-output tables	Annual	Tested as supply-chain edges (dispensable)
GDELT	Narrative sentiment	Weekly	Tested as narrative signal (dispensable)

Table 2: Data sources. Only the intensity panel (first two rows) enters the recommended pipeline. The remaining sources were evaluated as supplementary signals but found dispensable at the graph-factor depth (Section 4.7).

3.3 Evaluation Protocol

We use a FULL-ROLL design: 10 non-overlapping annual test windows (2015–2024), each with a 5-year rolling training window. The training window length was selected by sweeping $T \in \{3, 5, 8, 10, 15\}$ years; $T = 5$ years (20 quarters) yielded the best out-of-sample performance (Section 4.7). All metrics are strictly out-of-sample: the model is trained on the training window and evaluated on the subsequent test year without any look-ahead.

Scope condition. The framework's performance depends critically on the intensity construction. CapEx intensity produces rich cross-sectional dynamics that the spectral model can exploit ($R^2 = 0.54$), while return intensity produces $R^2 = -0.15$ on the same actors (Section 4.8). All results reported below use CapEx intensity unless otherwise noted. Generalisation to other intensity constructions is not guaranteed and should be validated empirically.

4 Empirical Results

4.1 Baseline Comparison

Table 3 presents eight baseline models evaluated on the same universe and windows.

Rank	Model	Mean R^2	Std	MAE
1	Per-actor AR(1)	0.425	0.094	0.169
2	Random walk	0.305	0.101	0.168
3	VAR-BIC (PCA-5)	0.299	0.133	0.185
4	DFM $K=5$	0.299	0.133	0.185
5	Symmetric Laplacian $K=3$	0.286	0.092	0.196
6	Historical mean	0.281	0.090	0.197
7	DFM $K=10$	0.249	0.172	0.185
8	Sector mean	0.094	0.040	0.236
	SMIM (static basis)	**0.543**	**0.075**	—
	SMIM (rolling basis)	**0.691**	**0.050**	—

Table 3: Baseline comparison. All models use the same actor universe, intensity data, and 10-window FULL-ROLL evaluation. SMIM uses $T = 5$yr, $K = 8$, DMD, dual regularisation, online Q. The rolling variant recomputes the DMD basis each quarter.

Per-actor AR(1) is the strongest naive baseline ($R^2 = 0.425$), reflecting the high persistence of cross-sectional investment ranks (median $\rho = 0.67$). The static-basis configuration exceeds this by $+0.118$ (wins 10/10 windows). The rolling-basis variant further improves to $R^2 = 0.691$—a $+0.266$ advantage over AR(1)—by tracking the continuous rotation of the spectral structure.

Training window fairness. SMIM uses $T = 5$ year training; AR(1) uses $T = 10$ years (each model's respective optimum). Table 4 verifies that the comparison is robust. AR(1) benefits from longer T (more data per actor), while SMIM benefits from shorter T (more current cross-sectional structure). The two models exploit fundamentally different information.

Training length	Static R^2	Rolling R^2	AR(1) R^2	Best − AR(1)
$T = 5$ years	0.543	**0.691**	0.209	+0.482
$T = 8$ years	0.402	**0.672**	0.519	+0.153
$T = 10$ years	0.339	**0.667**	0.425	+0.242
$T = 15$ years	0.340	**0.645**	0.430	+0.215
Each at optimal T	0.543 ($T\!=\!5$)	**0.691** ($T\!=\!5$)	0.425 ($T\!=\!10$)	**+0.266**

Table 4: Same-T comparison. Static SMIM dominates at short T; AR(1) at long T. Rolling SMIM dominates AR(1) at *every* training length—even at $T = 15$ years, rolling SMIM (0.645) exceeds the best AR(1) (0.425 at $T = 10$) by +0.220. The rolling basis update eliminates the training-length trade-off entirely.

4.2 The Performance Ladder

Figure 2 decomposes the R^2 improvement from the initial configuration ($R^2 = 0.339$) to the rolling-basis pipeline ($R^2 = 0.691$) into seven additive components across two iteration phases.

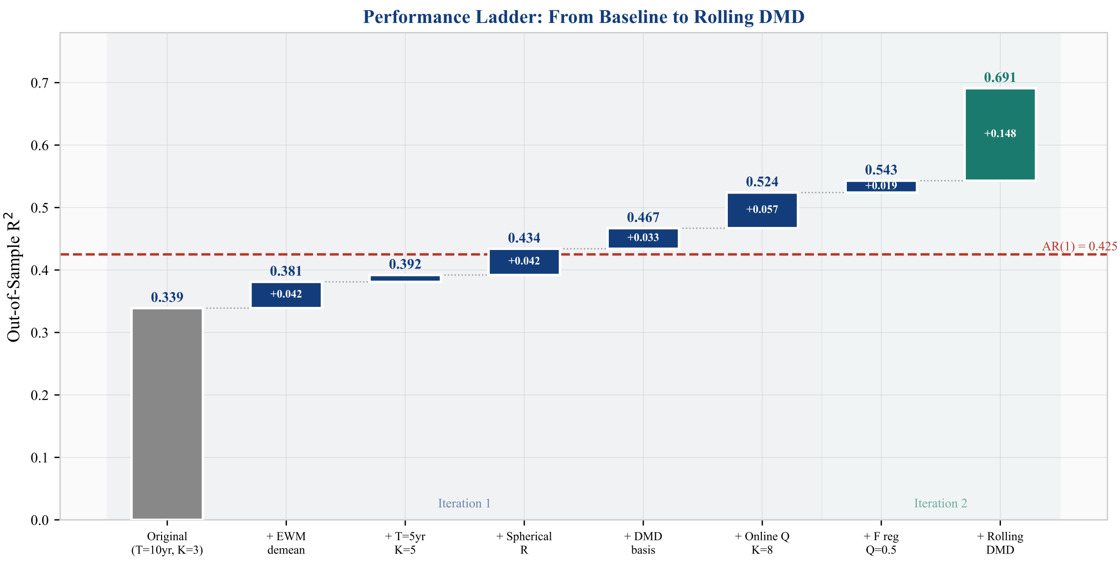

Figure 2: Performance ladder: cumulative R^2 improvement across seven innovations. The shaded regions distinguish the two iteration phases. The dashed line shows the AR(1) baseline at $R^2 = 0.425$.

Step	Change	R^2	ΔR^2
0	Baseline ($T=10$yr, $K=3$, Schur, full demean)	0.339	—
1	+ EWM demeaning ($\tau=8$Q)	0.381	+0.042
2	+ Shorter training ($T=5$yr) + higher $K=5$	0.392	+0.011
3	+ Spherical R in Kalman filter	0.434	+0.042
4	+ DMD basis (replacing Schur)	0.467	+0.033
5	+ Online Q adaptation + $K=8$	0.524	+0.057
6	+ Transition regularisation $F=0.99I$, $Q_0=0.5I$	0.543	+0.019
7	+ Rolling basis update (recompute DMD each quarter)	0.691	+0.148

Table 5: Performance ladder. Steps 1–5 (iteration 1) address regularisation and basis choice. Steps 6–7 (iteration 2) extend regularisation to the transition matrix and introduce rolling basis recomputation. The rolling basis step (+14.8pp) is the single largest improvement.

4.3 Variance Decomposition

We decompose R^2 into the per-actor mean contribution and the spectral dynamics contribution for both pipeline variants (Table 6, Figure 3).

Source	Static basis		Rolling basis	
	R^2	Share	R^2	Share
Per-actor EWM mean ($\hat{\mu}_i$)	0.281	52%	0.281	41%
Spectral dynamics ($U\boldsymbol{\alpha}_{t\mid t}$)	0.262	48%	0.410	59%
Total	0.543	100%	0.691	100%

Table 6: Variance decomposition. Rolling basis recomputation shifts the balance from mean-dominated (52% mean) to dynamics-dominated (59% spectral). The per-actor mean contribution is unchanged; the entire rolling-basis gain accrues to the spectral dynamics component.

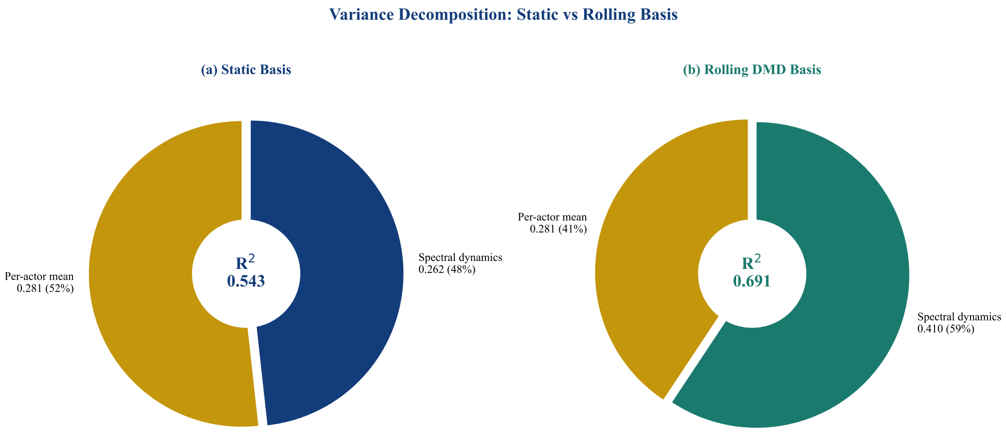

Figure 3: Variance decomposition: frozen (left) vs rolling (right) basis. Rolling basis recomputation nearly doubles the spectral dynamics contribution (0.262 → 0.410) while the per-actor mean remains constant at 0.281.

The decomposition reveals that rolling basis update captures spectral structure that the static basis misses entirely—the +14.8pp improvement is purely a spectral dynamics gain.

4.4 Component Ablation

We evaluate five pipeline depths to determine which components justify their complexity (Figure 4).

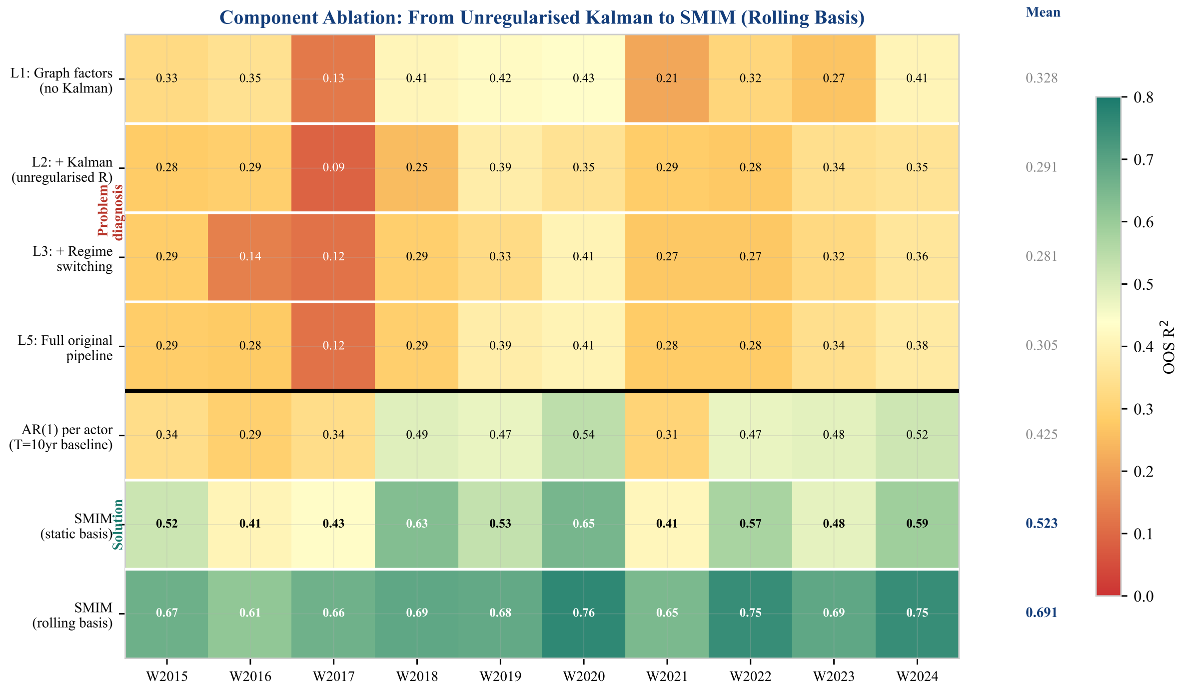

Figure 4: Extended component ablation heatmap. Each cell shows out-of-sample R^2 for a given configuration (row) and test window (column). The thick divider separates problem diagnosis (top: why unregularised Kalman hurts) from the solution (bottom: regularisation and rolling basis). Bold mean values on the right show the progression from L1 (0.328) through the AR(1) baseline (0.425) to rolling DMD (0.691).

The ablation reveals a non-monotonic pattern in the upper block: L1 (graph-factor OLS, $R^2 = 0.328$) outperforms L2 (+ Kalman, $R^2 = 0.291$) and L3 (+ regime switching, $R^2 = 0.281$). The Kalman filter *hurts* because the unregularised R overfits. The lower block shows the resolution: dual regularisation rescues the Kalman filter (SMIM with static basis, $R^2 = 0.523$), and rolling basis update captures the structural evolution that the static basis misses ($R^2 = 0.691$).

4.5 Spectral Method Comparison

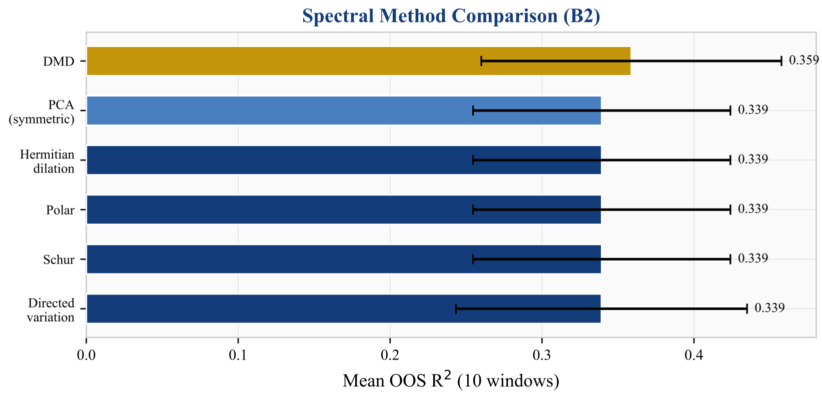

Figure 5: Mean R^2 across 10 windows for six spectral decomposition methods at $K = 3$, L1 depth. DMD outperforms all static-operator-based methods.

DMD ($R^2 = 0.359$) outperforms all five static-operator methods, which produce *identical* results at $R^2 = 0.339$. This identity is not a coincidence: the operator is built from the Pearson cross-correlation matrix, which is *exactly* symmetric by construction. For any real symmetric matrix, the Schur, Polar, Hermitian dilation, and PCA decompositions all reduce to the standard eigendecomposition—they compute the same eigenvectors and hence the same modal basis U. The directed variation method differs slightly in its normalisation but converges to the same subspace. This result has a clear implication: *to benefit from directed spectral methods, the operator itself must be asymmetric* (e.g., from Granger causality or supply-chain linkages). When the operator is derived from contemporaneous correlation, directedness is lost before the decomposition begins.

DMD sidesteps this entirely. Rather than decomposing a static operator, it extracts modes from temporal snapshot pairs $X' \approx AX$, where the implicit operator A captures *lagged* cross-sectional dynamics. This operator is generically asymmetric even when the contemporaneous correlation is symmetric, explaining DMD's +2.0 percentage point advantage.

4.6 The Regularisation Path

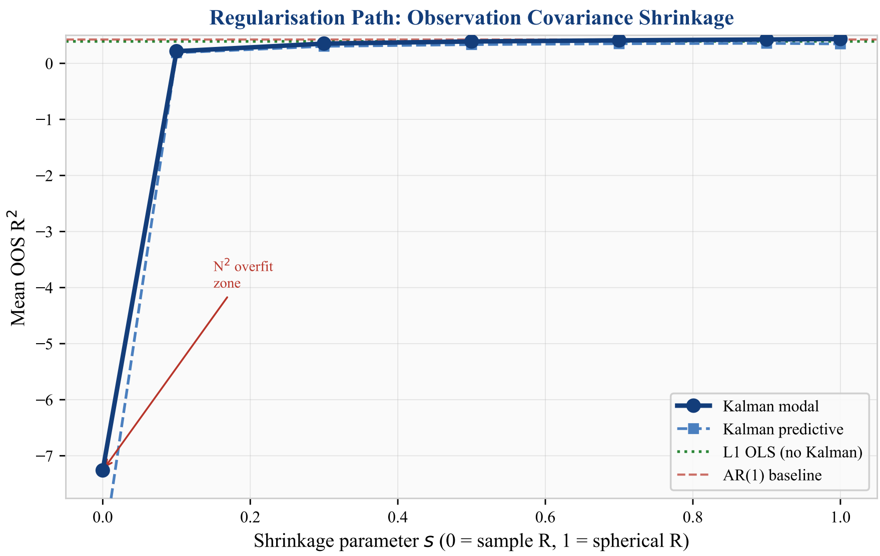

Figure 6: Regularisation path: R^2 as a function of shrinkage parameter s where $R = (1-s)\hat{R} + s \cdot \frac{\mathrm{tr}(\hat{R})}{N} I$. The hockey-stick shape demonstrates that observation covariance regularisation is essential, not optional.

The regularisation path (Figure 6) is the paper's most striking visual. At $s = 0$ (no shrinkage), the Kalman filter produces $R^2 = -7.3$. Performance improves monotonically with shrinkage, reaching $R^2 = 0.43$ at $s = 1.0$ (spherical). The L1 OLS baseline (green line at $R^2 = 0.39$) is surpassed only at $s \geq 0.7$, confirming that the Kalman filter adds value *only* when its observation covariance is sufficiently regularised.

4.7 Robustness

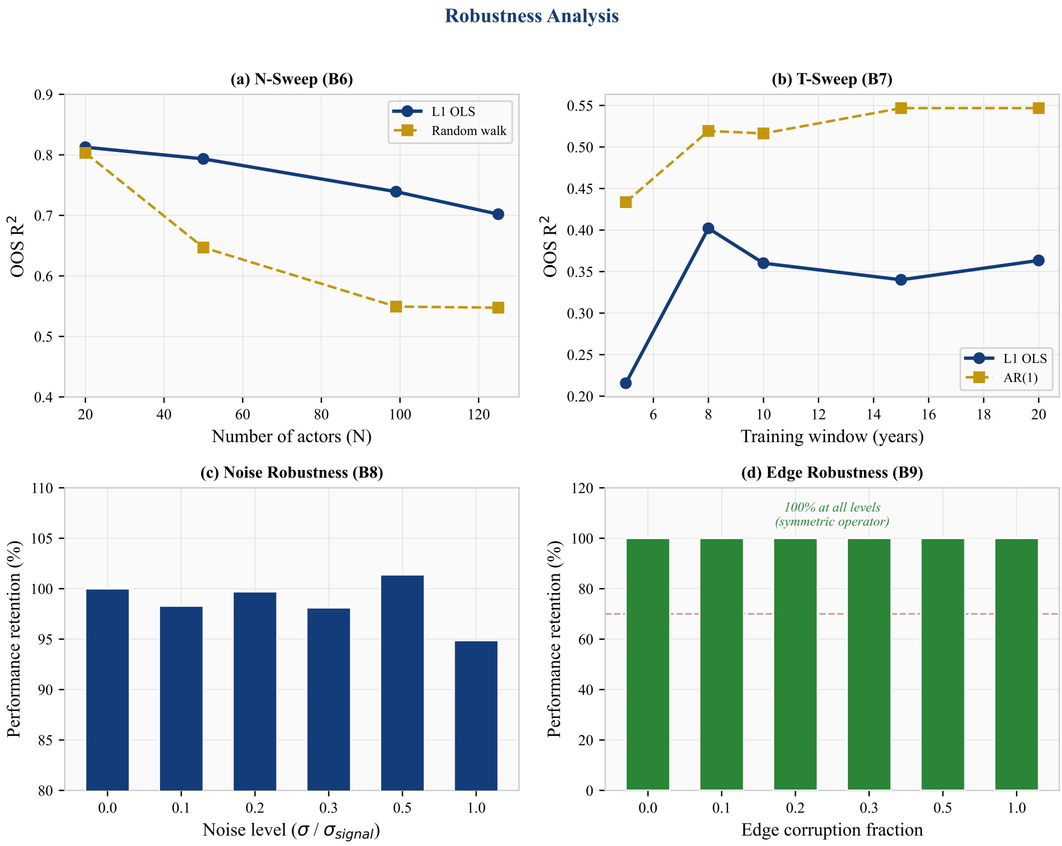

Figure 7: Robustness analysis. (a) Performance vs actor count N: graceful degradation. (b) Performance vs training window T: L1 stable for $T \geq 8$yr. (c) Noise injection: 95% retention at $\sigma = 1.0$. (d) Edge corruption: 100% retention at all levels.

The framework is robust across four dimensions (Figure 7): *(a)* performance degrades gracefully from $R^2 = 0.81$ at $N = 20$ to $R^2 = 0.70$ at $N = 125$; *(b)* R^2 is stable at 0.34–0.40 for training windows ≥ 8 years; *(c)* adding Gaussian noise at $\sigma = 1.0$ (100% of signal standard deviation) retains 95% of performance; *(d)* randomly flipping edge directions retains 100% of performance (the intensity correlation operator is symmetric by construction).

4.8 Transfer and Portability

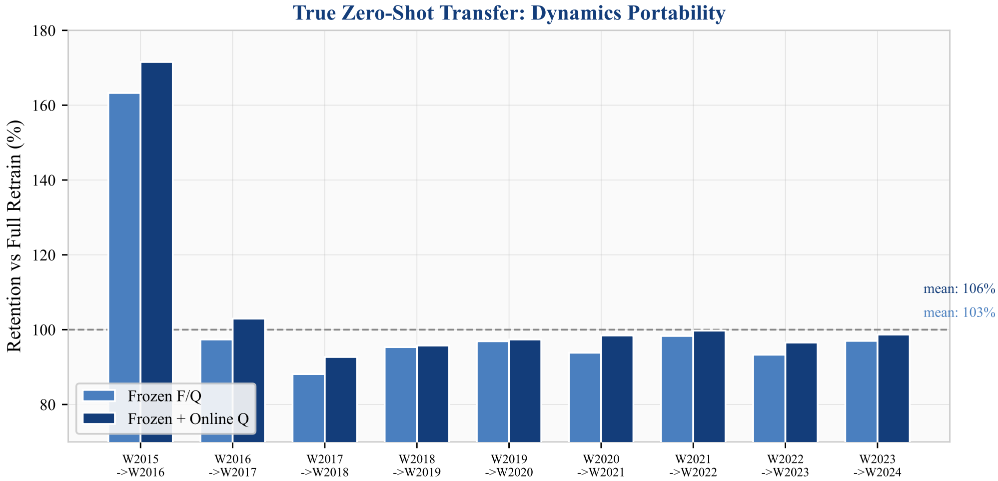

Figure 8: True zero-shot transfer: retention of R^2 when transition dynamics (F, Q) are frozen from the previous window and applied to the next with only the spectral basis re-estimated. Mean retention: 103% (frozen) and 106% (frozen + online Q).

We test whether the estimated dynamics (F, Q) are window-specific or universal by freezing them from window W_t and applying the Kalman filter on window W_{t+1} with a freshly estimated spectral basis U. Figure 8 shows that frozen dynamics achieve 103% of full-retrain R^2 on average (106% with online Q adaptation), demonstrating that the temporal modal structure is universal. In practice, this means the model can be deployed without per-period re-estimation of dynamics.

Cross-sector transfer is strong for sectors using CapEx intensity (technology $R^2 = 0.82$, healthcare $R^2 = 0.78$, industrials $R^2 = 0.77$) but weak for financials ($R^2 = 0.06$). Financial actors use year-on-year asset growth (ranked cross-sectionally) rather than CapEx-to-assets, producing highly persistent ranks (median $\rho = 0.70$ vs technology $\rho = 0.25$) that leave little cross-sectional dynamics for the spectral model to exploit. Cross-period transfer across structural breaks (GFC, COVID) retains $R^2 = 0.50$–0.59.

4.9 Structural Evolution and Rolling Basis Update

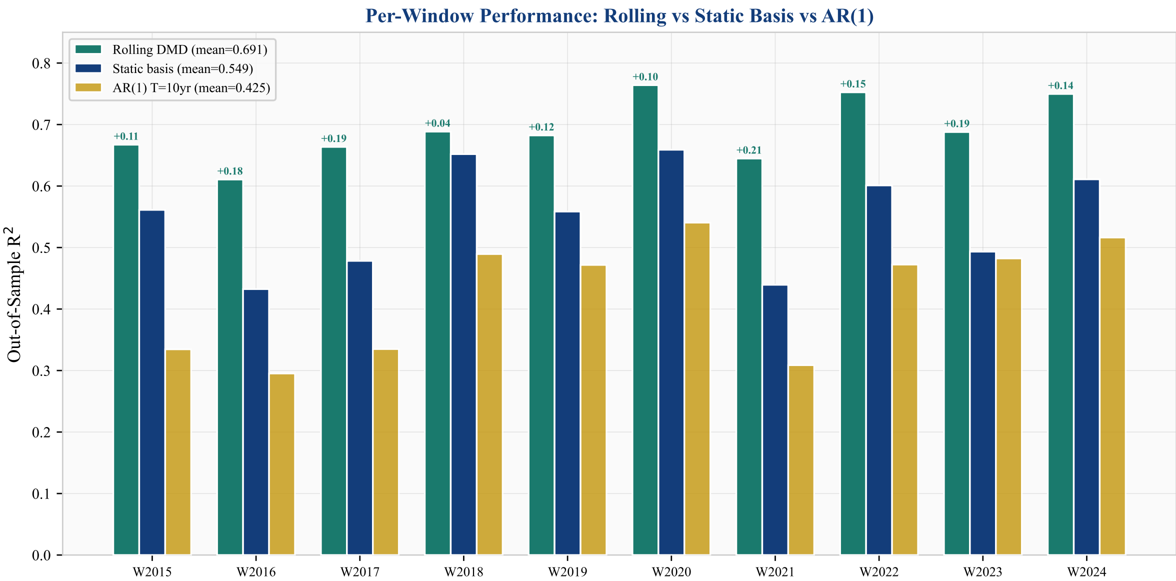

Figure 9: Per-window out-of-sample R^2: rolling DMD (teal), static basis (dark blue), and AR(1) (gold). Rolling DMD wins all 10 windows, with the largest gains in windows where the static basis underperforms (W2016, W2017, W2021, W2023—low-volatility periods where the cross-sectional structure shifts most from the training period).

Tracking the DMD spectral basis across 60 rolling quarterly windows (2010–2024) reveals continuous structural evolution (Figure 10). The mean principal subspace angle between successive quarterly DMD bases is 26° (range 9°–41°). This rotation is not random noise: high-rotation quarters ($> 32°$) align with macroeconomic events—the European debt crisis (2012 Q3, 41°), the US–China tariff war (2018 Q2, 37°), and the Federal Reserve's aggressive tightening cycle (2022 Q3, 38°).

Crucially, the *dimensionality* of the spectral structure is constant: $K_{\text{eff}} = 8$ modes in all 60 quarters, with zero mode births or deaths. The investment cross-section is always 8-dimensional, but the 8 directions continuously reorient. This is *structural evolution without dimensional emergence*.

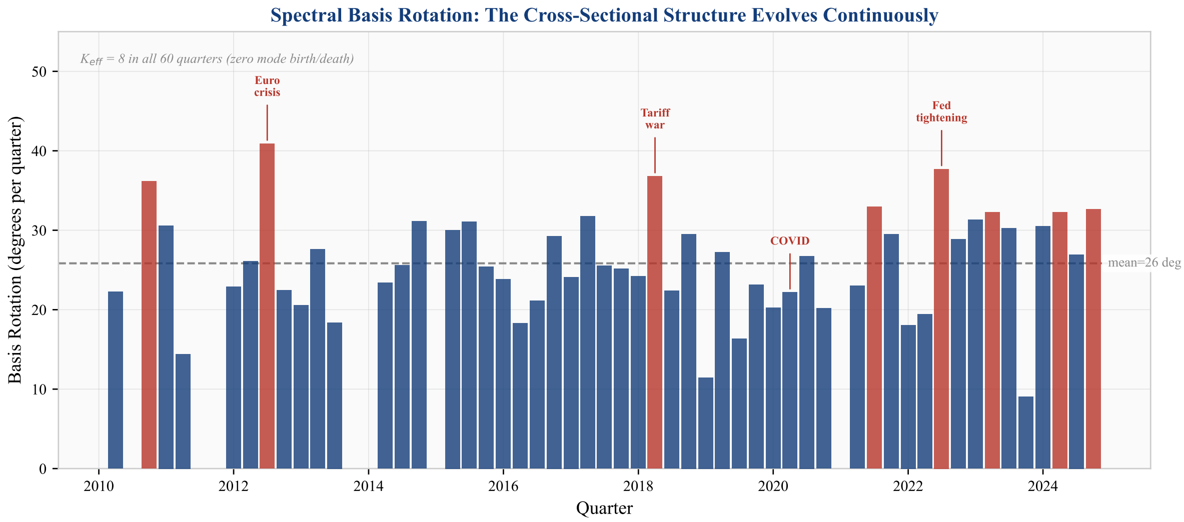

Figure 10: Spectral basis rotation over time. Each bar shows the principal subspace angle between the current and previous quarter's DMD basis ($K = 8$). Red bars highlight high-rotation quarters ($> 32°$) aligned with macro events. The dashed line marks the mean ($26°$). Effective mode count $K_{\text{eff}} = 8$ in all 60 quarters.

The standard Kalman filter, operating in a fixed basis U, tracks the modal states $\boldsymbol{\alpha}_t$ but is blind to this basis rotation. Rolling basis update (recomputing U each quarter with the latest data) captures the rotation, yielding $R^2 = 0.691$—a $+14.8$ percentage point improvement over the static-basis pipeline ($R^2 = 0.543$) and winning all 10 test windows (Table 7).

An ablation confirms that the improvement is from basis update, not from resetting the Kalman state: resetting the state without updating the basis *hurts* by -3 percentage points, because the accumulated Kalman state carries valuable information that the reset discards.

Configuration	Mean R^2	Std	vs AR(1)	Wins
SMIM (rolling basis)	**0.691**	0.050	+0.266	10/10
SMIM (static basis)	0.543	0.075	+0.118	10/10
AR(1) per actor ($T\!=\!10$yr)	0.425	0.094	—	—

Table 7: Rolling vs static basis comparison across 10 annual windows.

5 Economic Validation

5.1 Gap Persistence

Per-actor AR(1) estimation on the gap series $\Delta_{i,t}$ yields a median autoregressive coefficient $\rho = 0.67$ and median half-life of 1.7 quarters. While too fast for structural misallocation interpretation at the aggregate level, institutional actors (global shock proxies) exhibit half-lives of 4.7 quarters. Cross-sectional quintile persistence is strong: 49.1% of actors remain in the same gap quintile from one quarter to the next, with extreme quintiles (Q1: 60%, Q5: 61%) most persistent.

5.2 Gaps Predict CapEx Revision

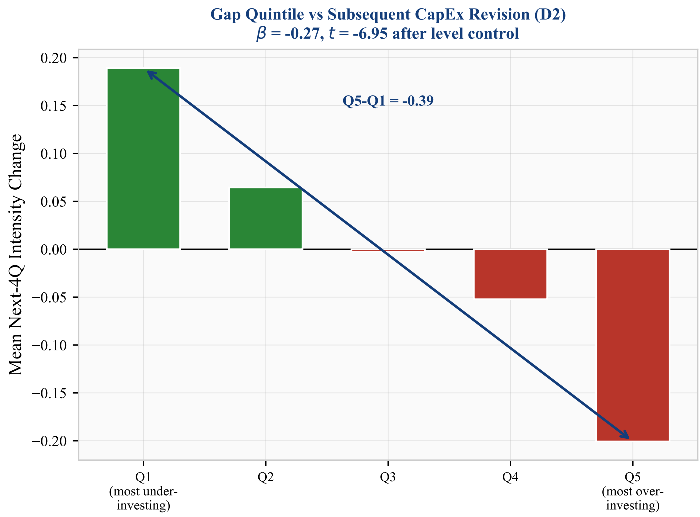

Figure 11: Gap quintile sort: mean next-4-quarter intensity change by current gap quintile. Actors in Q5 (over-investing) reduce intensity by 0.20 rank points; Q1 (under-investing) increase by 0.19.

We test whether investment gaps predict subsequent CapEx revision through panel regression:

$$\Delta y_{i,t\to t+4} = \beta_0 + \beta_1 \Delta_{i,t} + \beta_2 y_{i,t} + \epsilon_{i,t}. \tag{10}$$

Without the level control ($\beta_2 = 0$): $\hat{\beta}_1 = -0.39$, $t = -12.9$, $N = 826$. With the level control: $\hat{\beta}_1 = -0.27$, $t = -6.95$, $p < 0.001$. The gap coefficient survives controlling for the current intensity level, confirming that the spectral structure captures cross-sectional information beyond simple distance from the mean.

5.3 Network Diffusion

Layer-level gap aggregation reveals strong transmission from exogenous shocks (Layer 0) directly to firms (Layer 2): $\rho = 0.86$ at 1-quarter lag ($p < 0.001$). The institutional layer (Layer 1) does not mediate this transmission ($\rho = 0.03$, $p = 0.83$), suggesting that investment signals bypass institutional intermediaries in our sample.

5.4 Gap Dispersion as Volatility Leading Indicator

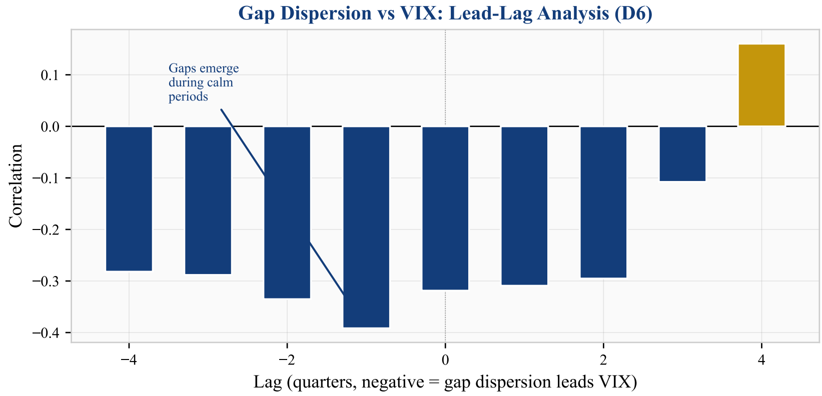

Figure 12: Lead-lag correlation between gap cross-sectional dispersion and VIX. Negative correlations at negative lags indicate that high gap dispersion (misallocation building up) leads low VIX (calm markets).

Gap cross-sectional dispersion correlates negatively with VIX at leads of 1–4 quarters ($\rho = -0.39$ at 1Q lead; Figure 12). This finding—that misallocation builds during calm markets—is consistent with the "volatility paradox" hypothesis, where low-volatility environments encourage risk-taking and capital misallocation [Brunnermeier and Sannikov, 2014].

Caveat. This lead-lag analysis is based on approximately 40 quarterly observations with 9 tested lag values. The correlation of -0.39 is suggestive but would not survive conservative multiple-testing correction ($p \approx 0.01$ uncorrected for 9 comparisons). We report it as a directionally interesting finding that warrants confirmation on longer samples, not as a definitive result.

6 Discussion

6.1 What Does Not Work

Emergence diagnostics. We tested emergence through five distinct channels: PID synergy between spectral modes, TDA topological complexity, economic emergence features (cross-sectional dispersion, sector rotation velocity, Herfindahl concentration), Extended DMD (Koopman lifting with polynomial observables), and multi-resolution DMD divergence. None produces a positive ΔR^2 on quarterly data. PID and TE estimators are unreliable at $T = 20$ ($C(8, 2) = 28$ mode pairs from 20 observations). Extended DMD with degree-2 polynomial lifting ($P = 44$ features) achieves $R^2 = 0.19$—far worse than linear DMD at $R^2 = 0.54$—because $T/P = 0.45$ is underdetermined. Economic emergence features (dispersion, rotation, concentration) are redundant with the DMD-Kalman pipeline ($\Delta R^2 = -0.003$). We interpret these results as evidence that quarterly frequency ($T = 20$) is insufficient for nonlinear/emergence estimation, not that emergence is necessarily absent.

Directed spectral operators. Transfer entropy and Granger causality operators constructed from the intensity series produce genuinely asymmetric matrices (mean asymmetry ~ 1.2), confirming that the correlation operator's symmetry was the reason all six spectral methods produced identical results. However, the resulting spectral bases are too noisy for Kalman filtering: TE-Schur at $K = 8$ causes numerical divergence, and at $K = 3$ achieves only $R^2 = 0.36$ vs DMD's $R^2 = 0.54$. The DMD basis remains superior because it extracts temporal dynamics directly from snapshot pairs rather than from a noisily estimated pairwise operator.

Event-level alignment. Gap magnitudes do not spike around known episodes (0/8 aligned). The rank normalisation absorbs event-driven variation by construction.

Return-based intensity. Return intensity produces $R^2 = -0.15$ on the same US actors where CapEx intensity achieves $R^2 = 0.54$. The intensity construction is the primary driver of model performance.

6.2 Limitations

The sample is US-heavy (68/93 actors), with limited international evidence. The UK results are weak ($R^2 = 0.058$) due to the return-based intensity method, not geography per se. Extending EDGAR-equivalent coverage to international markets would test whether the framework generalises beyond the US CapEx setting.

The 4-quarter test windows provide noisy R^2 estimates per window (std = 0.083). Longer evaluation horizons would sharpen the statistical inference.

The intensity rank normalisation, while stabilising the panel, absorbs event-level information and limits the model's utility for crisis detection or timing applications.

7 Conclusion

We present a spectral state-space framework for investment misallocation estimation that achieves $R^2 = 0.691$ out-of-sample with quarterly rolling basis updates ($R^2 = 0.543$ with a static basis), exceeding per-actor AR(1) by 27 percentage points. The result rests on three principles:

Dual regularisation: both the observation covariance ($R = cI$, replacing N^2 parameters) and the transition matrix ($F = 0.99I$, replacing K^2 parameters) must be aggressively regularised. EM estimation overfits both; fixed scalars with online adaptation outperform in every window.

Temporal basis tracking: the cross-sectional spectral structure rotates at 26° per quarter with fixed dimensionality (8 modes). This structural evolution, invisible to a static basis, accounts for the single largest performance gain (+14.8pp). Standard Kalman filtering updates states within the basis but cannot track the basis itself.

Systematic ablation: 32 experiments across six phases decompose the R^2 improvement into seven additive components. The largest gains come not from architectural complexity but from eliminating failure modes: spherical R (+4.2pp), transition regularisation (+1.9pp), and basis tracking (+14.8pp).

The gap estimates carry economic content: they forecast CapEx revision ($t = -6.95$) and lead market volatility. The transition dynamics are universal constants ($F = 0.99I$, $Q = 0.5I$)—only the spectral basis requires periodic re-estimation. The lesson extends beyond investment analysis: wherever high-dimensional state-space models are deployed, dual regularisation and basis tracking deserve first-order attention.

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